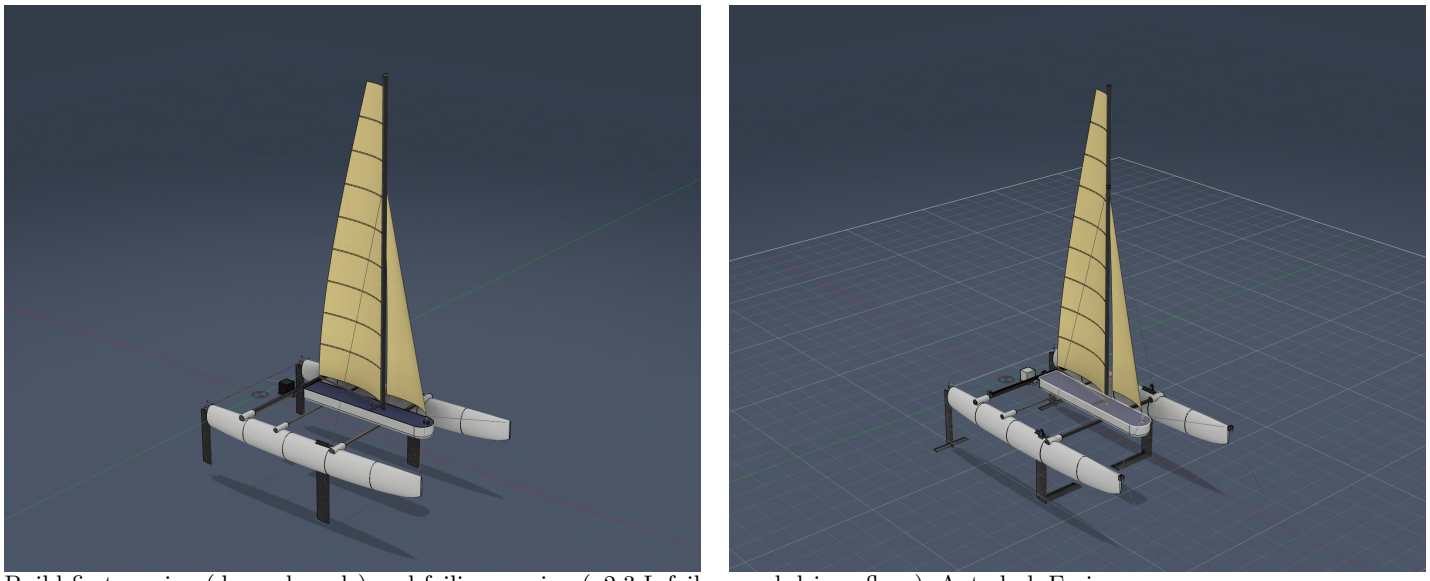
A 1 m RC Foiling Beach Catamaran: Design, Analysis and Build Plan

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Build-first version (daggerboards) and foiling version (v2.3 L-foils, wand-driven flaps), Autodesk Fusion.

Goal

A radio-controlled catamaran that follows full-size beach cats (Hobie 16, Nacra 17), can be 3D-printed and built from off-the-shelf RC parts, can fly on hydrofoils, and whose behaviour is predicted before it is built.

What was done

- **Parametric CAD** in Autodesk Fusion, driven through its API: 1 m hulls printed as sealed PETG segments, a beach-cat rig trimmed by three sliders, two appendage options, a 0.6 m first build, clearance sweeps, and mass and inertia exported for analysis.
- **Velocity prediction**: ITTC-57 friction, Michell thin-ship wave resistance with catamaran interference (checked on the Wigley hull), foil and appendage drag, and sails eased for heel and bow-down limits.
- **Foils**: lifting-line wings with wand-driven flaps; flying trim in heave and pitch, roll authority, coupled heave–pitch stability, and a time-domain take-off simulation.
- **Flight simulator** (web page, same physics) and a search of 306,180 foil designs under limits on stress, deflection, Reynolds number, pitch stability and damping, with depth margins and robustness checks.
- **Build plan**: every fitting drawn and printable (tracks and cars, gudgeons, wing-mast fairing, gooseneck, ride-height linkage); costed parts lists (every price read from a retailer page), build manuals for the 0.6 m and 1 m boats, and a tow-test plan for the foils.

Findings

- Fixed-incidence foils have no heave stability; a wand-driven flap on each foil makes heave and pitch stable (least damping ratio 0.69).
- The first optimized foils (v2.1) would not have flown: at speed the sail drive's bow-down couple makes the elevators push down, and at $+4°$ they pitch the boat nose-down until they leave the water. The adopted v2.3 foils set the elevators at $-0.5°$ and fly level.
- Near take-off the foils have no roll authority left, so flying at $60°$ needs 8.9 kn against 7.1 kn on a beam reach; taking off on a reach and heading up keeps it flying at $60°$ from 8.2 kn.
- Predicted speed on a beam reach: build-first 2.65 m/s in 8 kn; foiling 4.53 m/s in 12 kn.

Deliverables

Technical paper, A0 poster, build manual with 1:1 sail patterns, tow-test plan, interactive simulator, STEP/Fusion models and print-ready STLs. Checks in the code: 28 of 28 foil rules and 28 of 28 design rules pass.

Cost

0.6 m first build $257.37; 1 m build-first boat $381.02; foiling kit $101.17 (September 2026 retailer prices, before shipping and tax).

Next

Build and GPS-log the 0.6 m boat and compare with its predicted 2.1 m/s in 8 kn; build the 1 m boat; tow-test the foils against the plan's predictions, then fly.

All performance figures are model predictions; nothing has been tested on the water yet. Developed with Claude (Anthropic) as an AI engineering assistant for the CAD scripting, analysis code and documents.